

EXPERIMENT NO 17

Determination and plotting of Antenna Gain Calculation.

Aim:

To determine and plot Antenna Gain, and calculate the antenna gain in dBi

Software required:

- SCILAB version 6.0.1

Program:

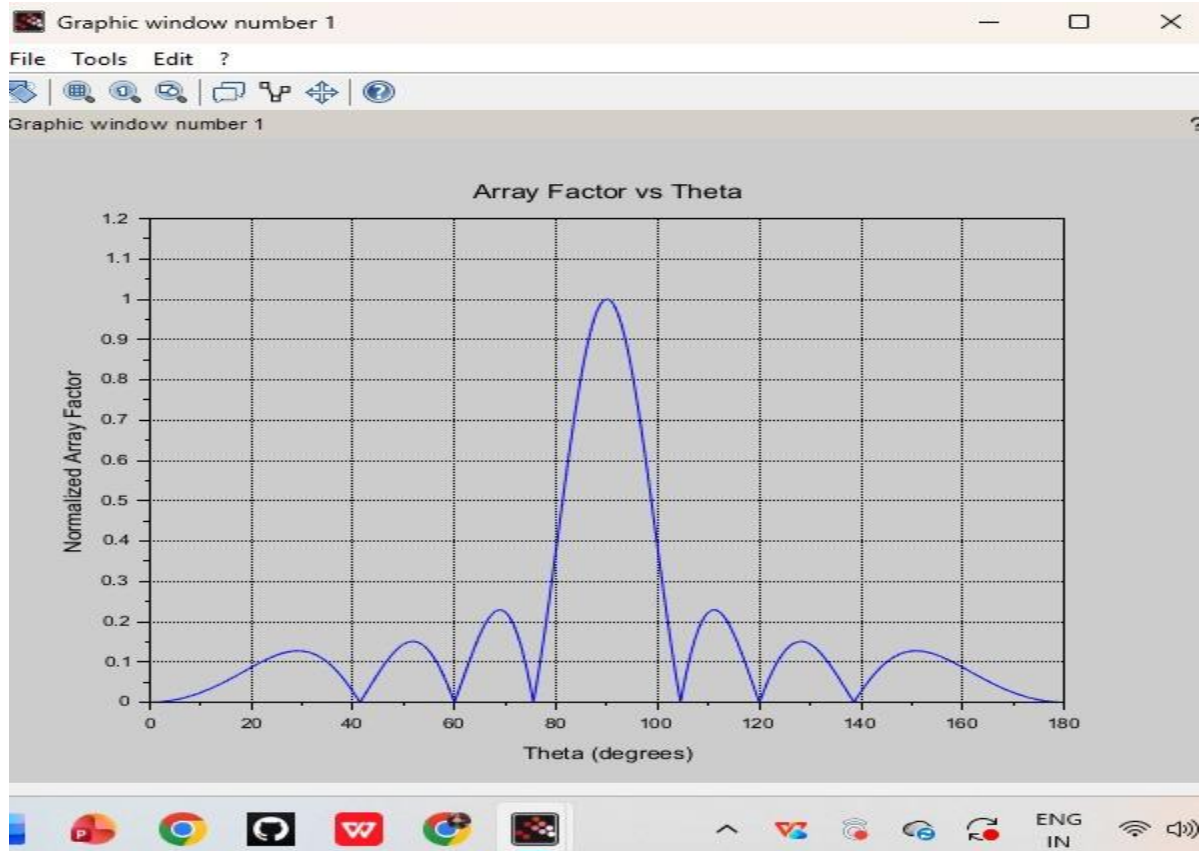
```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci X
1 clc;
2 clear;
3 clf();
4
5 // =====
6 // ARRAY FACTOR VS THETA
7 // =====
8
9 // Antenna parameters
10 N = 8; // Number of elements
11 d = 0.5; // Element spacing in wavelength
12 beta = 0; // Progressive phase shift
13
14 // Angular range: 0 to 180 degrees
15 theta_deg = linspace(0, 180, 2000);
16 theta = theta_deg * %pi / 180;
17
18 // =====
19 // ARRAY FACTOR CALCULATION
20 // =====
21
22 // Since d = lambda/2:
23 // k*d = pi
24
25 psi = %pi * d / 0.5 * cos(theta) + beta;
26
27 // Initialise array factor
28 AF = zeros(1, length(theta));
29
30 for i = 1:length(theta)
31
32     denominator = sin(psi(i) / 2);
33
34     if abs(denominator) < 1e-10 then
35         AF(i) = 1;
36     else
37         AF(i) = abs(sin(N * psi(i) / 2) / ...
38                 (N * denominator));
39     end
40
41 end
```

```

40
41 end
42
43 //=====
44 //NORMALIZATION
45 //=====
46
47 AF = AF / max(AF);
48
49
50 //=====
51 //PLOT
52 //=====
53
54 figure(1);
55
56 plot(theta_deg, AF);
57
58 xlabel("Theta- (degrees)");
59 ylabel("Normalised Array Factor");
60 title("Array Factor vs Theta");
61
62 xgrid();
63
64 //Set axis limits
65 a = gca();
66 a.data_bounds = [0, 0, 180, 1.05];
67

```

Output:



RESULT: The experiment was done and the gain calculation for the given antenna was done. Different case studies were analyzed and the gain was calculated.